

MUHAMMAD IRFAN RUSYDI BIN RAZALEIGH

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SUMMARY

Junior Game Programmer with expertise in C++, Unreal Engine, and XR development. Skilled in applying Object-Oriented Programming (OOP), Entity Component Systems (ECS), and 3D mathematics to build complex game logic, and AI-driven NPCs.

WORK EXPERIENCE

Passion Republic | Petaling Jaya, Selangor

Sept 2026 – Now

Junior Game Engineer

- Ensuring that the game design is fully implemented and performs as expected.
- Designing and implementing efficient and maintainable code.

CTRM Aero Composites, DRB HICOM | Batu Berendam, Melaka

Oct 2025 – Mar 2026

Design Engineering Intern | XR / Digital Twin Developer

- Built a VR-based digital twin of CTRM's forklift simulation and mission-based training system using C++ in Unreal Engine.
- Integrated real-world hardware, including Logitech wheel and Arduino controls, into VR environments.
- Implemented Mixed Reality using Meta Quest 3 with passthrough and hand tracking.
- Managed the 3D pipeline from Blender to Unreal Engine and executed VR deployment (APK).

PROJECTS

Dynamic AI-Driven NPCs with Real-Time Adaptive Conversations | Unreal Engine

Jun – Aug 2025

- AI-driven NPCs with real-time text-to-speech and lip-sync using Unreal Engine 5, Python, and C++.
- Integrated local LLM (Ollama) with NVIDIA Riva TTS for NPC dialogue and audio playback.
- Automated facial animation via Audio2Face for real-time lip-sync and expression.

Blackjack | C++ & SFML

Jul 2023

- Developed a digital adaptation of the classic casino card game Blackjack using C++, SFML, and Visual Studio 2022.
- Created a single-player mode versus an AI dealer featuring SFML-based 2D graphics.

CERTIFICATIONS

- **Certificate of Language Proficiency (English)** - CEFR C2 (goFLUENT, May 2026)
- **Unity Certified User: Programmer** (Unity Technologies, Sep 2025)
- **Math & Logic:** Introduction to Linear Algebra with MATLAB, MATLAB Onramp

EDUCATION

Universiti Teknikal Malaysia Melaka

Ayer Keroh, Melaka

Bachelor of Information Technology (Game Technology)

Oct 2022 - Mar 2026

- **Cumulative GPA:** 3.7/4.0; Dean's List Oct 2022 – Aug 2025
- **Awards (GDIID UTeM 2025):** Gold Award, Industry Award, Best Presenter

SKILLS

- **Programming Languages:** C, C++, C#, Python, JavaScript (with HTML, CSS)
- **Tools & Deployment:** Unreal Engine 5, Unity, Blender, Git/Github, Diversion, Visual Studio
- **Graphic APIs & Libraries:** OpenGL, Raylib, SFML, Three.js
- **Databases:** phpMyAdmin, MySQL
- **Concepts & Practices:** Gameplay Programming, Entity Component Systems (ECS), Object-Oriented Programming (OOP), 3D Mathematics (Linear Algebra), Test Driven Development (TDD), Software Development Life Cycle (SDLC)
- **Spoken Languages:** Malay (Native), English (Fluent), Mandarin (Beginner)

ACTIVITIES

Interactive Media Club, FTMK

Ayer Keroh, Melaka

President of Club

Jun 2025 – Jun 2025

- Led and co-led 7+ events, including Industrial Visit to Larian Studios, Visual Effect Workshop, IMC Let's Meet Up, Industrial Visit to Warnakala Studios, and Game Development Workshop.
- Administered event planning and budgeting, coordinating 11 cross-courses team members with 95% on-schedule execution.